



TEAM VYGOTSKY



REFRESHER

GENED Mathematics



LET & CSE
#1 ONLINE REVIEW AUTHORITY



Gurong Pinoy





PRAYER

Dear Lord, I come to you to ask for your guidance and direction in this study session. I ask that the holy spirit fill me with strength, creativity and understanding to get through my studies without difficulty or sin. Help me hold my focus and attention. Help me to retain all that I learned. Please make my mind sharp and keep distractions at bay. If any part of my studying is weak or lacking in some way, let it be revealed so that I may correct it. Thank you for this opportunity to learn. Amen.

**MATH ka lang...
Magiging**

LPT

AKO!!



1. One-way fare from Manila to Cebu is ₱3,800. How many round-trip tickets can be purchased with the amount of ₱91,200?

a. 8

b. 14

c. 10

☒ d. 12

₱3,800 = one-way fare

₱3,800 X 2 = ₱7,600 - cost of 1 round-trip ticket

₱91,200 / ₱7,600 = 12 round trip tickets

2. A water tank contains 18 liters when it is 20% full. How many liters does it contain when 50% full?

a. 60

$$P = 18$$

b. 30

$$R = 20\% \text{ or } 0.20$$

c. 58

$$B = ?$$

d. 45

$$B = P/R$$

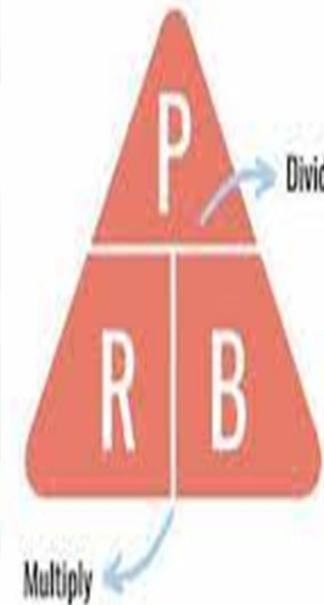
$$B = 18/0.2$$

$$B = 90 \text{ liters when full}$$

When 50% full:

$$= 90 \times 0.5$$

$$= 45 \text{ liters}$$



Tandaan ang mga sumusunod na formulas!

To find the **Percentage (P)**:
 $P = \text{Rate (R)} \times \text{Base (B)}$

To find the **Rate (R)**:
 $R = \text{Percentage (P)} \div \text{Base (B)}$

To find the **Base (B)**:
 $B = \text{Percentage (P)} \div \text{Rate (R)}$

3. Which of the following has the greatest value?

a. $3 + 3^2 + (3 + 3)^2$

b. 3^3

c. $(3 + 3 + 3)^2$

d. $[(3 + 3)^2]^3$

$$3 + 3^2 + (3 + 3)^2 = 3 + 9 + 36 = 48$$

$$3^3 = 27$$

$$(3 + 3 + 3)^2 = 9^2 = 81$$

$$[(3 + 3)^2]^3 = \{(6)^2\}^3 = (36)^3 = 46656$$

PEMDAS

PEMDAS is an acronym used to remember the order of operations: parentheses, exponents, multiplication/division, addition/subtraction

P

Parentheses

E

Exponents

MD

Multiplication/Division
(left to right)

AS

Addition/Subtraction
(left to right)


$$(5 + 8)^2 \times 6 \div 3 + 12 - 4$$

4. The average of five counting numbers is 20. What is the highest possible value that one of the numbers can have?

- a. 20
- b. 30
- c. 40
- ☒ d. 90

Let: 1, 2, 3, 4 - the first four numbers

N - the highest number

20 - average of the five numbers

Formula of Average



$$\text{average} = \frac{\text{sum of values}}{\text{number of values}}$$

$$20 = \frac{1+2+3+4+N}{5}$$

$$5\left(20 = \frac{1+2+3+4+N}{5}\right)5$$

$$100 = 10 + N$$

$$100 - 10 = 10 - 10 + N$$

$$90 = N$$

5. Thirty students had an average of 75, while 20 students had an average of 90. What is the average of the entire group?

- a. 81
- b. 82
- c. 82.5
- d. 81.5

$$\text{Ave} = \frac{(30)(75) + (20)(90)}{50}$$

$$\text{Ave} = \frac{2,250 + 1,800}{50}$$

$$\text{Ave} = \frac{4,050}{50}$$

$$\text{Ave} = 81$$

Average Formula

$$\text{Average} = \frac{(a_1 + a_2 + \dots + a_n)}{n}$$



6. How much greater is the sum of the first 100 counting numbers than the sum of the first 50 counting numbers?

- a. 110
- ✓ b. 3775
- c. 3177
- d. 1200

$$S_{n(50)} = \left(\frac{50}{2}\right) (1 + 50)$$

$$S_{n(50)} = 25 \times 51$$

$$S_{n(50)} = 1,275$$

$$S_{n(100)} = \left(\frac{100}{2}\right) (1 + 100)$$

$$S_{n(100)} = 50 \times 101$$

$$S_{n(100)} = 5,050$$

$$S_{n(100)} - S_{n(50)} = 5,050 - 1,275$$

$$S_{n(100)} - S_{n(50)} = 3,775$$

Formula

Sum of N terms of an
Arithmetic Series

$$S_n = \frac{n}{2} (a_1 + a_n)$$

a_1 : First term in the sequence.

a_n : Last term in the sequence.

n : Number of terms in the sequence.

7. If a man uses 20 gallons of gasoline to travel 350 km in his car, how many gallons will he need for a 750 km trip?

- a. 42.86
- b. 52.86
- c. 62.86
- d. 72.86

$$20 : 350 = X : 750$$

$$15,000 = 350X$$

$$15,000/350 = 350X/350$$

$$X = 42.857 \text{ or } 42.86$$



Product of extremes = Product of means
 $a \times d = b \times c$

8. Which of the following is the factorization of the binomial $x^2 - 4$?

a. $(x+4)(x-2)$

b. $(x+2)(x+2)$

c. $(x-2)(x-2)$

☒ d. $(x-2)(x+2)$

Difference of Squares Formula.



DIFFERENCE OF SQUARES FORMULA

$$a^2 - b^2 = (a + b)(a - b)$$

Check using the FOIL method

$$\begin{aligned} & x^2 - 4 \\ &= x^2 - 2^2 \\ &= (x+2)(x-2) \end{aligned}$$

$$\begin{aligned} &= (x-2)(x+2) \\ &= (x^2 + 2x - 2x - 4) \\ &= x^2 - 4 \end{aligned}$$

9. The vertex of an isosceles triangle is 20 degrees. What is the measure of one of the base angles?

a. 150 degrees

b. 75 degrees

✓ c. 80 degrees

d. 90 degrees

Let: x be measure of one base angle

$$x + x + 20 = 180$$

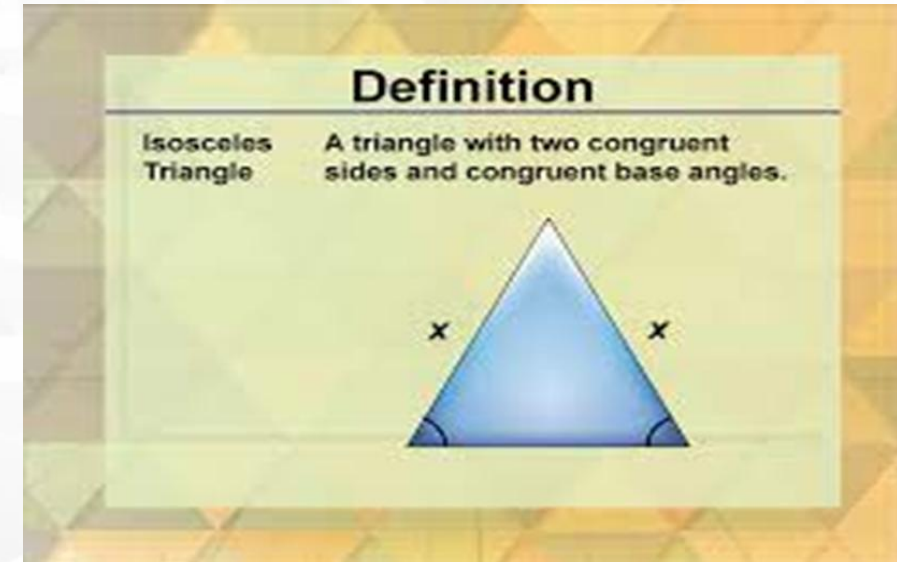
$$2x + 20 = 180$$

$$2x = 180 - 20$$

$$2x = 160$$

$$2x/2 = 160/2$$

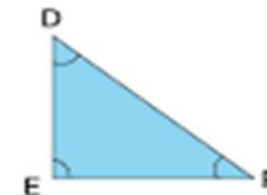
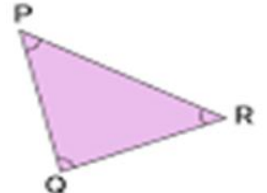
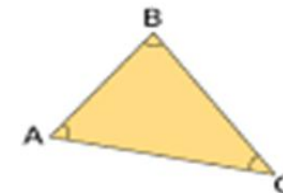
$$x = 80$$



Angle Sum Property of a Triangle



The angle sum property of a triangle states that the angles of a triangle always add up to 180°



$$\angle A + \angle B + \angle C = 180^\circ$$

$$\angle D + \angle E + \angle F = 180^\circ$$

$$\angle P + \angle Q + \angle R = 180^\circ$$

10. Only 26% of 50 students passed the exam. How many students failed the test?

✓ a. 37

B = 50 total no. of students

b. 35

R = 26% or 0.26 passed

c. 33

P = ?

d. 39

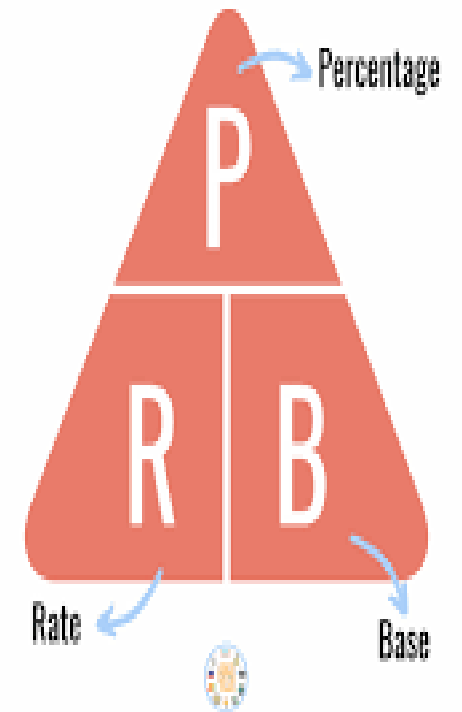
$P = R \times B$

$P = 0.26 \times 50$

P = 13 – no of students who passed

$$50 - 13 = 37$$

TECHAN'S TRIANGLE



11. How much commission does a salesman get if he was able to sell ₱35,000 worth of jewelry and he is given a 15% of it?

- a. ₱5,300
- b. ₱5,325
- ☒ c. ₱5,250
- d. ₱5,425

$$B = ₱35,000$$

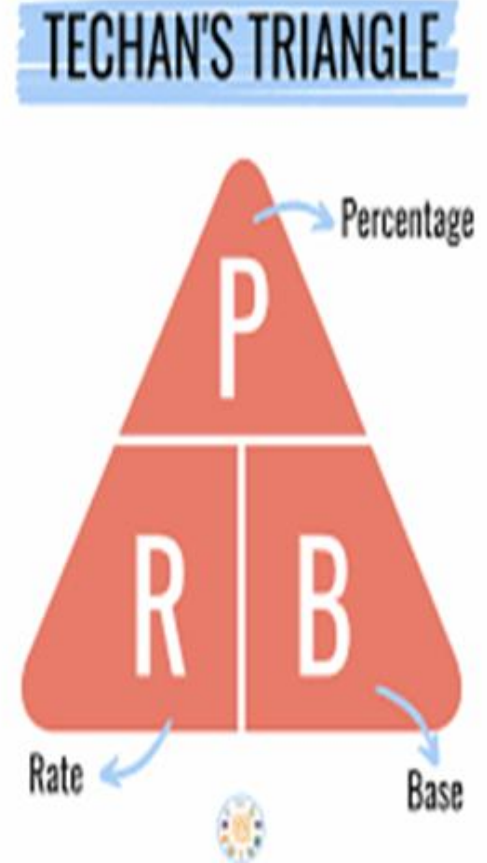
$$R = 15\% \text{ or } 0.15$$

$$P = ?$$

$$P = R \times B$$

$$P = 0.15 \times ₱35,000$$

$$P = ₱5,250$$



12. How much should a man deposit in a bank at 6% simple interest rate if he wants to earn ₱1,200 a year?

a. ₱10,000

b. ₱30,000

c. ₱15,000

☒ d. ₱20,000

$$I = ₱1,200$$

$$R = 6\% \text{ or } 0.06$$

$$T = 1 \text{ year}$$

$$P = ?$$

$$I = PRT$$

$$1,200 = P(0.06)(1)$$

$$1,200 = 0.06P$$

$$1,200/0.06 = 0.06P/0.06$$

$$20,000 = P$$

$$\text{Simple Interest} = P \times r \times t$$

- P → Principal
- r → Interest Rate
- t → Time in Years

13. Julius has ₱15,000 in the bank. After a year, he expects it to earn ₱750. What is the rate of interest?

- a. 2%
- b. 4%
- ☒ c. 5%
- d. 6%

$$P = \text{P}15,000$$

$$T = 1 \text{ year}$$

$$I = \text{P}750$$

$$R = ?$$

$$I = PRT$$

$$750 = (15,000) \times R \times 1$$

$$750 = 15,000R$$

$$750/15,000 = 15,000R/15,000$$

$$0.05 = R \text{ or } R = 5\%$$

Simple Interest
Formula

$$I = PRT$$

interest principal rate time

14. In a playground for kindergarten kids, 18 children are riding tricycles or bicycles. If there are 43 wheels in all, how many tricycles are there?

- a. 8
- b. 9
- ☒ c. 7
- d. 11

$$T + B = 18 \text{ ----- eq 1}$$

$$3T + 2B = 43 \text{ ----- eq 2}$$

Multiply eq 1 by -2 then add to eq 2

$$3T + 2B = 43 \text{ ----- eq 2}$$

$$-2T - 2B = -36$$

$$T = 7 \quad \text{No. of tricycles}$$

$$18 - 7 = 11 \quad \text{No. of bicycles}$$

To check:

$$3T + 2B = 43$$

$$3(7) + 2(11) = 43$$

$$21 + 22 = 43$$

$$43 = 43$$

15. How much will P25,000 amount to at 5% simple interest rate after 2 years?

- a. P27,500
- b. P28,500
- c. P2,500
- d. P27,000

$$P = \text{P}25,000$$

$$r = 5\% \text{ or } 0.05$$

$$t = 2 \text{ years}$$

$$A = ?$$

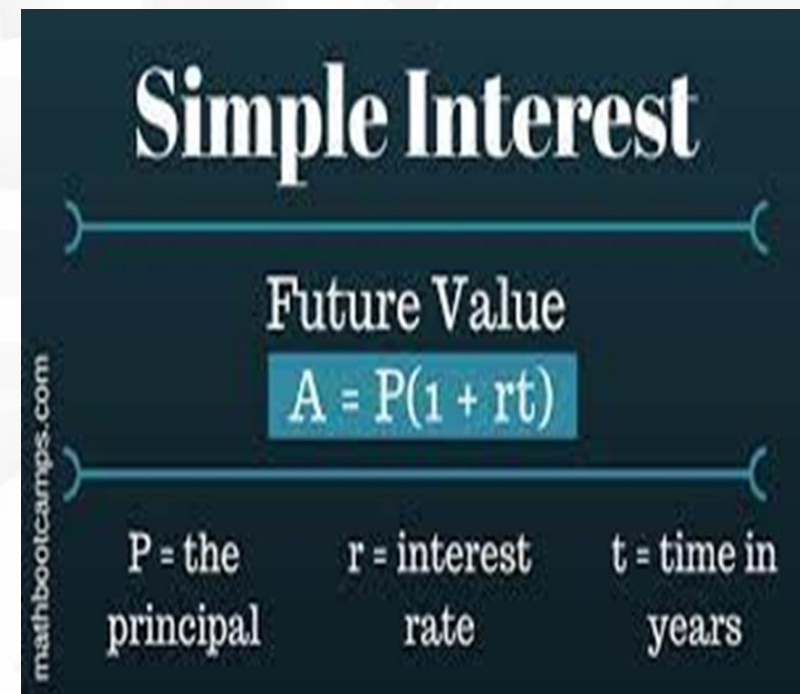
$$A = P(1 + rt)$$

$$A = 25,000[1 + (0.05)(2)]$$

$$A = 25,000(1 + 0.1)$$

$$A = 25,000 + 2500$$

$$A = \text{P}27,500$$



The diagram is titled "Simple Interest" in a large, bold, white serif font. Below the title, a horizontal line with curly braces at both ends spans the width of the diagram. Under this line, the words "Future Value" are written in a white serif font. Below "Future Value", the formula $A = P(1 + rt)$ is displayed in white text inside a light blue rectangular box. At the bottom of the diagram, there are three labels: "P = the principal", "r = interest rate", and "t = time in years", all in a white serif font. On the left side of the diagram, the text "mathbootcamps.com" is written vertically in a small, white, sans-serif font.

16. Which has the most volume?

- a. A box 10 meters high, 6 meters wide, and 14 meters long.
- b. A box 1 meter high, 60 meters wide, and 14 meters long.
- ✓ c. A box 9 meters high, 7 meters wide, and 14 meters long.
- d. A box 11 meters high, 8 meters wide, and 7 meters long.

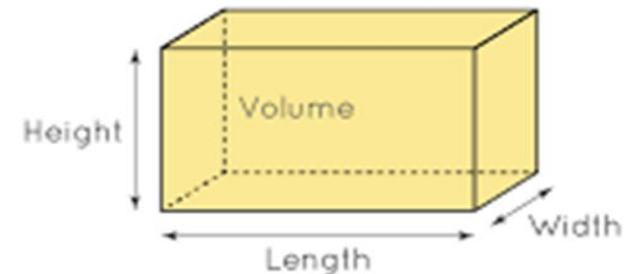
a. $10\text{m} \times 6\text{m} \times 14\text{m} = 840$ cubic meters

b. $1\text{m} \times 60\text{m} \times 14\text{m} = 840$ cubic meters

c. $9\text{m} \times 7\text{m} \times 14\text{m} = 882$ cubic meters

d. $9\text{m} \times 7\text{m} \times 14\text{m} = 616$ cubic meters

Volume of a Right
Rectangular Prism



Volume = length $\times$ width $\times$ height

17. The edges of a rectangular solid have these measures: 1.5 ft by 11/2 ft by 3 inches. What is its volume in cubic inches?

- a. 324
- b. 225
- ☒ c. 972
- d. 45

$$1 \text{ ft.} = 12 \text{ in.}$$

$$1.5 \text{ ft.} \times 12 \text{ in./ft} = 18$$

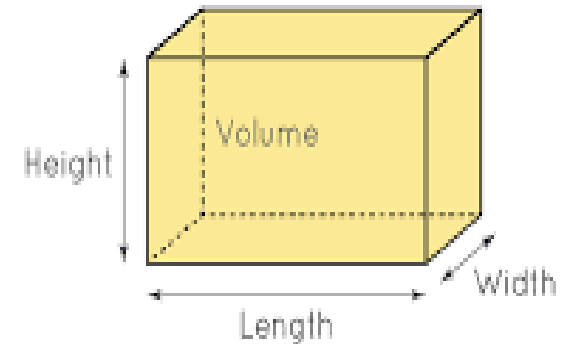
$$11/2 \text{ ft.} \times 12 \text{ in./ft} = 18$$

$$V = L \times W \times H$$

$$V = 18 \times 18 \times 3$$

$$V = 972 \text{ cu. in.}$$

Volume of a Right
Rectangular Prism



$$\text{Volume} = \text{length} \times \text{width} \times \text{height}$$

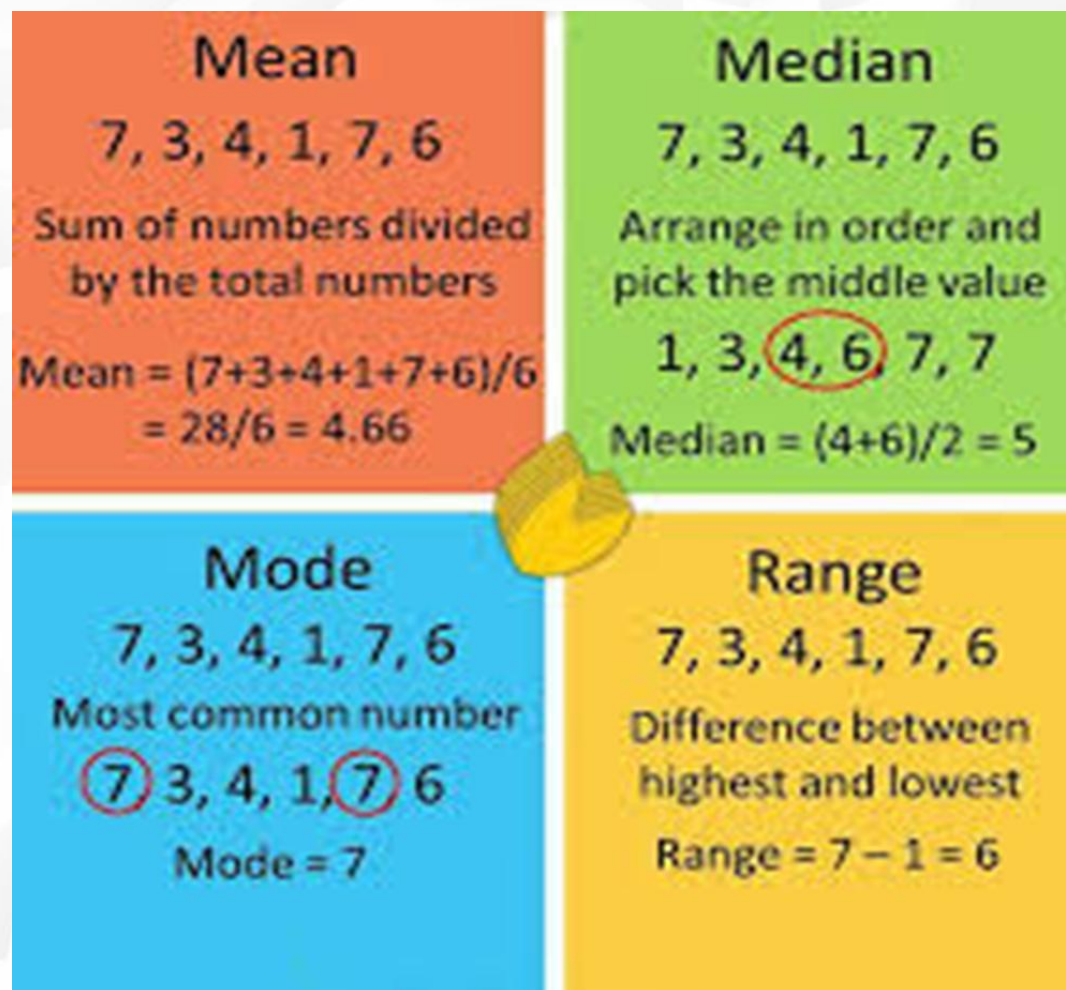
18. If the score of 10 students are: 76, 80, 75, 83, 80, 79, 85, 80, 88, 90, the mode is:

a. 85

b. 79

c. 88

☒ d. 80



19. Find the median of the following numbers 8, 5, 7, 5, 9, 9, 1, 8, and 10

a. 5

b. 7

☒ c. 8

d. 9

If n is odd

1, 5, 5, 7, 8, 8, 9, 9, 10

Median = 8

If n is even

1, 3, 5, 5, 7, 8, 8, 9, 9, 10

Median = $(7 + 8)/2 = 7.5$



Mean = $\frac{\text{Sum of all values}}{\text{Total number of all values}}$

Median = Middle value (when the data are arranged in order)

Mode = Most common value

20. Ruben's grades are 88, 90, 97, 90, 91 and 86. What is the grade that he should aim for in the 7th subject if he has to have an average of 91?

a. 97

☒ b. 95

c. 92

d. 89

$$91 = \frac{88+90+97+90+91+86+N}{7}$$

$$91 = \frac{542+N}{7}$$

$$7\left(91 = \frac{542+N}{7}\right)7$$

$$637 = 542 + N$$

$$637 - 542 = N$$

$$95 = N$$



Average
Formula

$$= \frac{\text{Total Sum of All Numbers}}{\text{Number of Item in the Set}}$$



21. If a certain job can be finished by 18 workers in 26 days, how many workers are needed to finish the job in 12 days?

a. 24

b. 30

☒ c. 39

d. 45

$$18:26 = x:12$$

$$18 (26) = 12 (x)$$

$$468 = 12x$$

$$468/12 = 12x/12$$

$$\mathbf{39 = x}$$

$$18 \times 26 = 39 \times 12$$

$$468 = 468$$

22. Three men are paid P315 to do a certain job. The sum is to be divided among them in the ratio of 4:5:6. How much does each receive?

- a. 84, 104, & 127
- ☒ b. 84, 105, & 126
- c. 83, 106, & 126
- d. 83, 105, & 127

$$4x + 5x + 6x = 315$$

$$15x = 315$$

$$15x/15 = 315/15$$

$$x = 21$$

$$4x = 4(21) = 84$$

$$5x = 5(21) = 105$$

$$6x = 6(21) = 126$$

23. The product of two whole numbers is 36, and their ratio is 1:4. Which of these is the smaller number?

- a. 9
- ☒ b. 3
- c. 2
- d. 12

Since the product of two numbers is 36, look for the factors of 36 that can satisfy the ratio 1:4

Factors of 36 :

$$1 \times 36 = 1 : 36$$

$$2 \times 18 = 1 : 9$$

$$3 \times 12 = 1 : 4$$

$$4 \times 9 = 4 : 9$$

$$6 \times 6 = 1 : 1$$

24. Three times the first of the three consecutive integers is three more than twice the third. Find the third integer.

✓ a. 9

b. 13

c. 11

d. 15

Let x – first integer

$x + 1$ – 2nd integer

$x + 2$ – 3rd integer

$$3x = 2(x + 2) + 3$$

$$3x = 2x + 4 + 3$$

$$3x = 2x + 7$$

$$3x - 2x = 7$$

$$x = 7 - 1^{\text{st}} \text{ integer}$$

$$x + 2 = 9 - 3^{\text{rd}} \text{ integer}$$

To check:

$$3 \times 7 = [2 \times 9] + 3$$

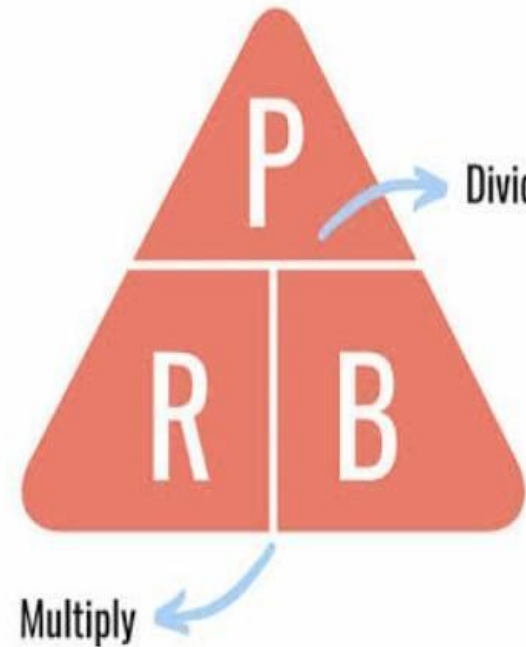
$$21 = 18 + 3$$

$$21 = 21$$

25. Julio's salary is ₱8,500 per month. If it is increased by 12%, what is his new salary per month?

- a. ₱9,320
- b. ₱9,420
- ☒ c. ₱9,520
- d. ₱9,550

$$\begin{aligned} &= ₱8,500 \times 0.12 \\ &= ₱1,020 \\ &= ₱8,500 + ₱1,020 \\ &= ₱9,520 \\ &\text{or} \\ &= ₱8,500 \times 1.12 \\ &= ₱9,520 \end{aligned}$$



Tandaan ang mga sumusunod na formulas!

To find the **Percentage (P)**:
 $P = \text{Rate (R)} \times \text{Base (B)}$

To find the **Rate (R)**:
 $R = \text{Percentage (P)} \div \text{Base (B)}$

To find the **Base (B)**:
 $B = \text{Percentage (P)} \div \text{Rate (R)}$

26. An article of clothing is on sale at 15% off the marked price. If the marked price is 1,050, what is the sale price?

a. 132.15

☒ b. 892.5

c. 157.5

d. 787.5

Marked price = 1050

Discount rate = 15%

Selling Price = ?

$$P = R \times B$$

$$P = 0.15 \times 1050$$

$$P = 157.5 - \text{Discount}$$

$$SP = LP - D$$

$$SP = 1050 - 157.5$$

$$SP = 892.5$$

Discount Formulas



$$\text{Discount} = \text{List Price} - \text{Selling Price}$$

$$\text{Discount \%} = \frac{(\text{List Price} - \text{Selling Price})}{\text{List Price}} \times 100$$

List Price :- The price in the label or tag of an item/product

Selling Price :- The price at which an item is sold

Discount % :- The amount of money reduced from the list price expressed as a percentage.

27. An elevator can carry a maximum load of 605 kg. How many passengers weighing 50.5 kg each can the elevator load?

- a. 10
- ☒ b. 11
- c. 12
- d. 13

$$605/50.5 = 11.98$$

11 persons only



28. Of the 50 students enrolled in the subject – Trigonometry, 90% took the final examination at the end of the term. Two-thirds of those who took the final exam passed. How many students passed the exam?

a. 33

b. 45

c. 34

☒ d. 30

$50 \times 0.9 = 45$ $\longrightarrow$ Took the final exam in Trigonometry

$45 \times \frac{2}{3} = 30$ $\longrightarrow$ Passed the exam

29. Of the following discounts, which is equal to the series discount 30%, 20%, 10%

- a. 40% $1 - d_1 = 1 - 0.3 = 0.7$
- b. 46.9% $1 - d_2 = 1 - 0.2 = 0.8$
- c. 48% $1 - d_3 = 1 - 0.1 = 0.9$
- ☒ d. 49.6%

$$\begin{aligned}SD &= 1 - (0.7)(0.8)(0.9) \\SD &= 1 - 0.504 \\SD &= 0.496 \text{ or } 49.6\%\end{aligned}$$

MULTIPLE
Discount Series

Single Discount $= 1 - [(1 - d_1)(1 - d_2)(1 - d_3)]$

30. How many times will the digit 7 appear between 1 to 100?

- a. 9
- ☒ b. 20
- c. 19
- d. 11

Whole Numbers 1 to 100

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

*You are
Stronger
than what you
Think*